25900653 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSESTA200 - STATISTICAL TECHNIQUES FOR DATA SCIENCE AND MACHINE LEARNING

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer type questions

Answer any **7** questions

Each question carries **2** marks

1. What are the assumptions of ANOVA of one way classified data ? [K] / [CO1]
2. Explain F-distribution. What are its applications? [K] / [CO1]
3. Give two examples of unsupervised learning used in daily life. [K] / [CO3]
4. Define linear regression. [K] / [CO3]
5. Define residuals in regression [K] / [CO3]
6. Name any two statistical tools used in Machine Learning. [K] / [CO3]
7. What is Time Series data? [U] / [CO4]
8. Mention one example of Data Science in transportation. [K] / [CO4]
9. Mention one feature of biological data. [K] / [CO4]
10. Define the impact of Data Science in simple terms. [K] / [CO4]

[2x7 = 14]

Part B

Short Essay Questions

Answer any **4** questions

Each question carries **6** marks

11. A random sample of rats fed on diet A over a period and another sample of rats fed on diet B over a period .We have got the following values: [A] / [CO1]

	Sample size	sample mean	sample standard deviation
Diet A	8	6	3.8
Diet B	5	9	4.15

Test whether diets A and B differ significantly in their variances.

12. Explain the test procedure for testing $H_0 : \mu = \mu_0$ against $H_1 : \mu < \mu_0$ when sample size $n < 30$ and σ is unknown. [U] / [CO1]

13. [A] / [CO2]

The following table gives the number of accidents that occurred during 7 days of the week. Find whether the accidents are uniformly distributed over the week.

Days	Sun	Mon	Tue	Wed	Thu	Fri	Sat
Number of accidents	14	18	15	11	15	14	14

14. From the following data on hair and eye colour can it be said that there is a significant association between hair colour and eye colour? [A] / [CO2]

		Hair Colour	
		Black	Brown
Eye Colour	Light	32	13
	Dark	14	21

15. Distinguish between traditional programming and Machine Learning. [U] / [CO3]

16. Discuss two real world-applications of structured and unstructured data. [U] / [CO4]

[6x4 = 24]

Part C

Essay Questions

Answer any **1** questions

Each question carries **12** marks

17. Evaluate the performance of a logistic regression model using a confusion matrix. Discuss metrics such as accuracy, precision, recall, and F1-score. [U] / [CO3]

18. Discuss in detail the evolution of data from traditional systems to the Big Data era. Highlight the different stages of data evolution with suitable examples. [U] / [CO4]

[12x1 = 12]